

Site Visit PLTP Ijen

28 Agustus 2025
PT PLN (Persero)



MEDCOENERGI
Geothermal

IJEN GEOTHERMAL POWER PLANT SUMMARY



Blawan Ijen Geothermal Power Plant Unit 1

The first geothermal power plant in East Java with a capacity of 35 MW, the plant is capable of supplying enough green energy to light up at least 75,000 households. The existence of the PLTP not only supports the stability of the eastern electricity system but also contributes significantly to the clean energy transition, sustainable geothermal utilization and local economy improvement

Net Capacity	: 35 MW (URC)
COD	: 09 February 2025
Plant Location	: Bondowoso, East Java
PPA Contract	: 110 MW



Well information:

- 6 Production Wells
- 1 Injection Well
- 1 Shut In Production Well (IJN 9-1 with ~7MW with less than 1 km distance)
- 1 Shut In Injection Well (IJN 2-1)



Our People:

- Exploration phase: 278 employees
- Development phase: 1,010 employees
- Operation phase: 86 employees



Power Transmission:

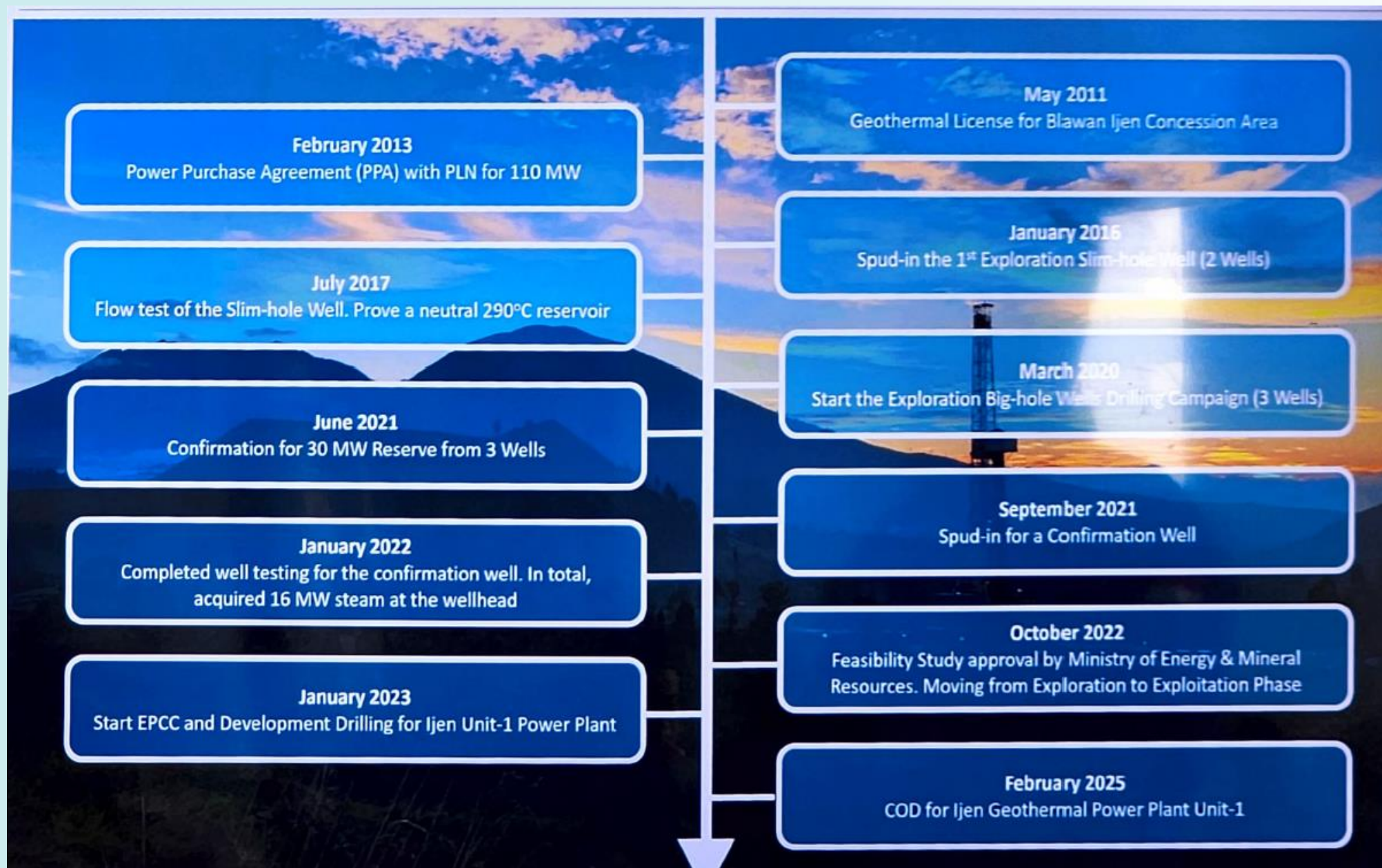
28 km for 150 kV Transmission Line (83 towers) from Power Plant to Banyuwangi Substation

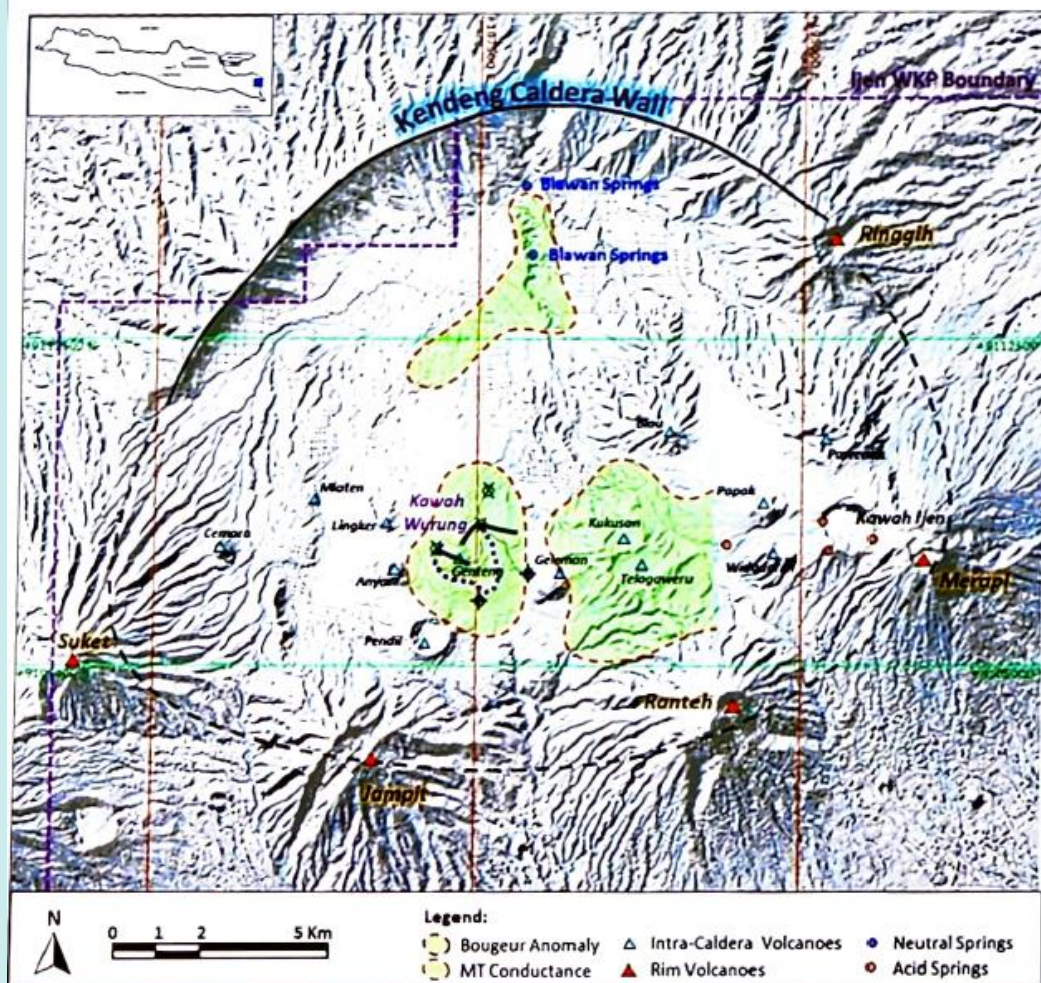


Achievements / Award:

- ✓ IBEA Award 2025 – Best IPP in Technology Innovation for Renewable Energy
- ✓ K3LL Award by East Java Governor (Jan 2025)
- ✓ Subroto Award by EBTKE (Dec 2024)
- ✓ Zero Accident Award by Ministry of Manpower (Jan 2024)
- ✓ ESG Award by PT SMI (Sept 2024)
- ✓ HIV-AIDS and Tuberculosis Prevention Award by East Java Government (2024)
- ✓ Subroto Award by EBTKE (2023)



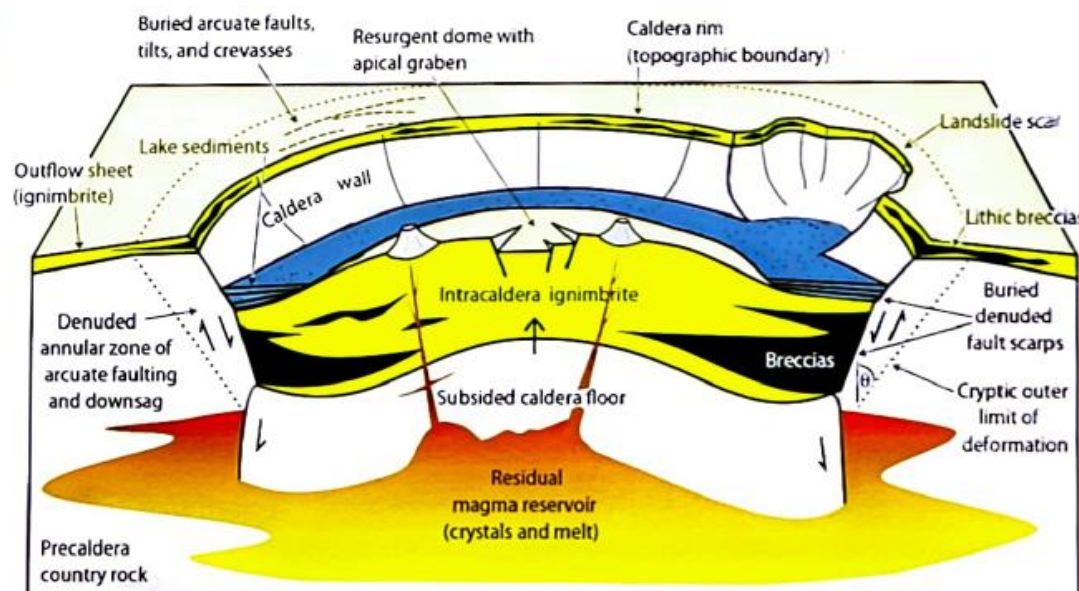


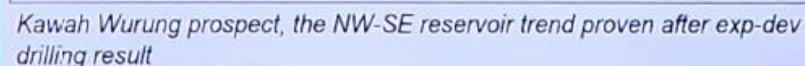


3 prospects in Kendeng Caldera:

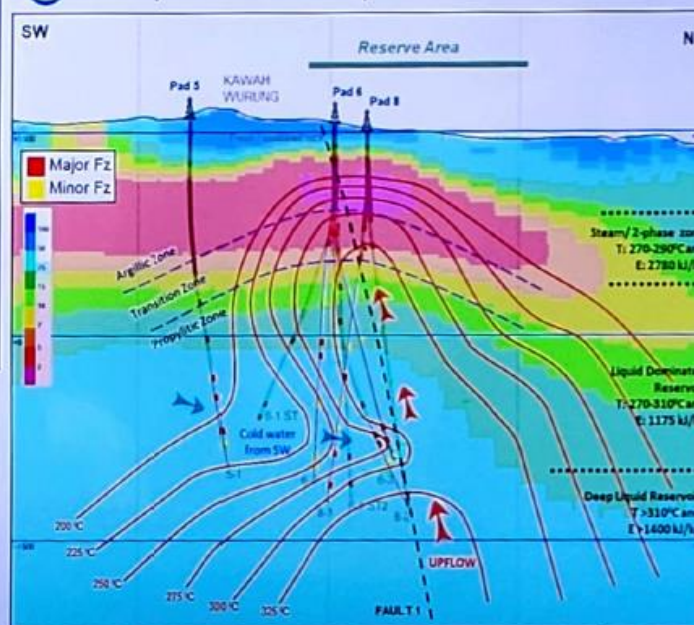
- 1) **Kawah Wurung** developed 35 MW (with possible additional 20-25 MW)
- 2) **Telaga Weru** prospect possible 45-50 MW
- 3) **Blawan** prospect intermediate system

- Geothermal Systems associated with heat source from caldera dome, intra-caldera volcanoes and rim volcanoes
- Manifestation limited at Kawah Ijen (acid springs) and Blawan (neutral thermal springs)
- One of the prospects is characterized as a “blind geothermal system”. Courageous and conservative exploration conducted, then, in February 2025, the first geothermal power plant of Ijen Unit-1 started the operation. The next units planned to be developed, utilizing a similar approach.
- Exploration drilling confirmed the existence of neutral geothermal reservoir, a thin steam cap on top of liquid dominated reservoir with temperature up to 325°C.





- ✓ 2 deep slim-hole exploration wells drilling in 2016-2017, IJN-01 (Pad 1) and IJN-02 (Pad 2), found 290°C T that led to Kawah Wurung Prospect
- ✓ 3+1 deep big-holes exploration wells drilling in 2020-2021, discovered high enthalpy liquid-dominated reservoir, neutral pH and 0.2 wt% TDS Na-Cl fluid and < 0.3 wt% dissolved NCG in reservoir. The thin steam / two-phase reservoir (270-290°C) on top of liquid dominated reservoir (290-324°C)
 - 3 wells from Pad 6 (acquired 2 production wells)
 - 1 well from Pad 5 (acquired injector)
- ✓ 6 deep big-holes development wells in 2023-2024
 - 1 well from Pad 2 (injector), 2 wells from pad 6 (producer), 2 wells from Pad 8 (producer), and 1 well from pad 9 (Back up producer)
- ✓ Complete steam requirement to start Unit 1 (35 MW) include 20% buffer



SW-NE conceptual model cross sections

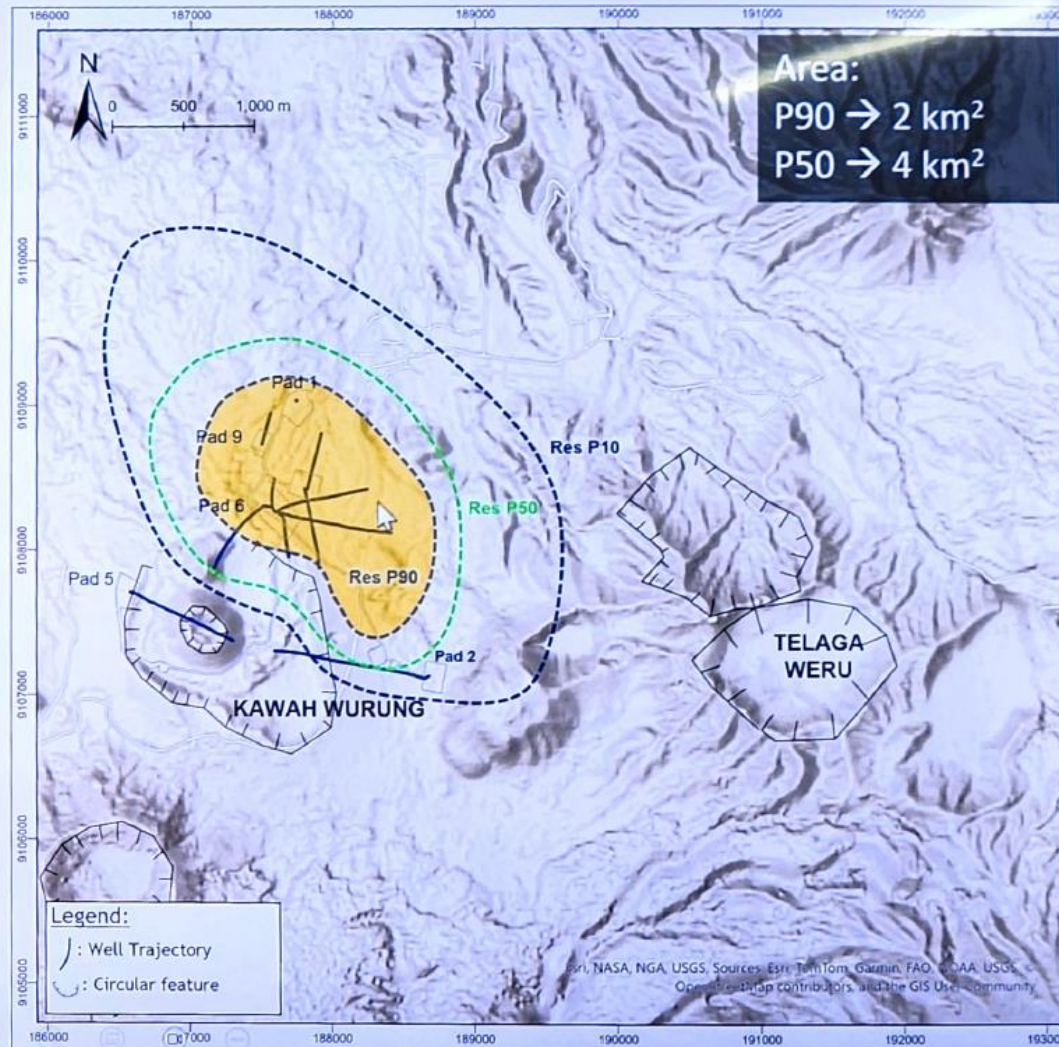


Simops drilling & flow test



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KAWAH WURUNG UNIT 1 DEVELOPMENT SUMMARY



Exploitation drilling sequence start with injection well at the south (IJN 2-1), as drilling and testing progress, required northern injection well (IJN 9-1). After drilled, IJN 9-1 well shows potential as production wells (high temperature, more than 320°C). IJN 9-1 proves productivity of the northern reservoir.

Power plant requirement with design of 1600 kJ/kg is requires ~120 kg/s total mass flow. Current operation of Unit 1 is supported by 6 production wells with total mass flow ~167 kg/s. Well IJN 9-1 on shut in condition and ready for next development.

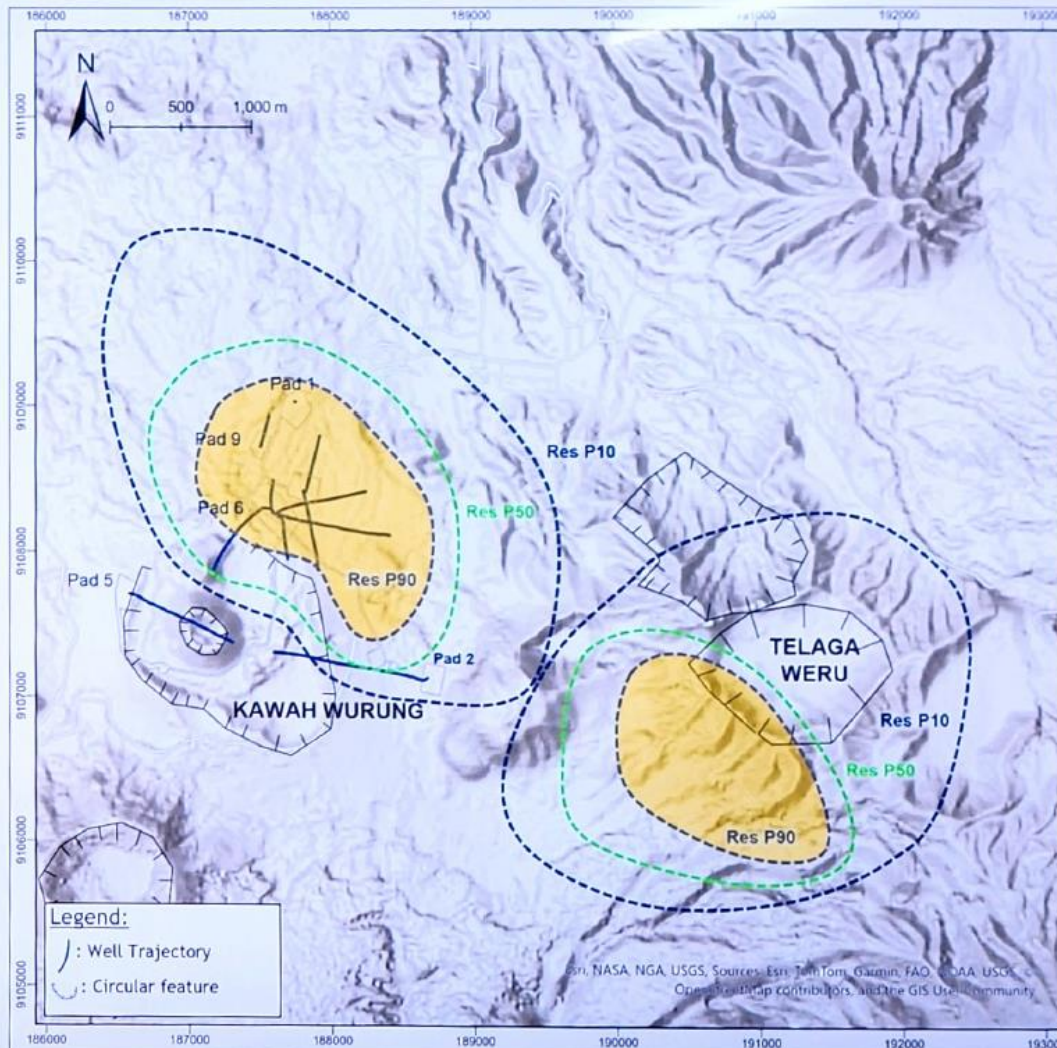
With power plant requirement of ~120 kg/s total mass flow, currently, injection capacity from IJN 5-1 is 158 kg/s and additional shut in injector well with capacity 95 kg/s from IJN 2-1.





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KAWAH WURUNG AND TELAGA WERU EXPANSION



Kawah Wurung Reservoir

- IJN 1 SH proven high temperature
- Available IJN 9-1 shut in well for production, proven high temperature and productive ~7 MW
- Ongoing monitoring performance of reservoir and well connectivity of Unit 1 for 6 months and will update Numerical Model and forecasting additional capacity of 20-25 MW

Telaga Weru Resources

- Telaga Weru appears mirror sector to Kawah Wurung, blind system with no direct surface manifestation
- Started the exploration, additional geoscience studies to fine tune previous geophysical (MT) data interpretation
- Resource estimation:
 - P50, 3 km²
 - P90, 1.5 km²



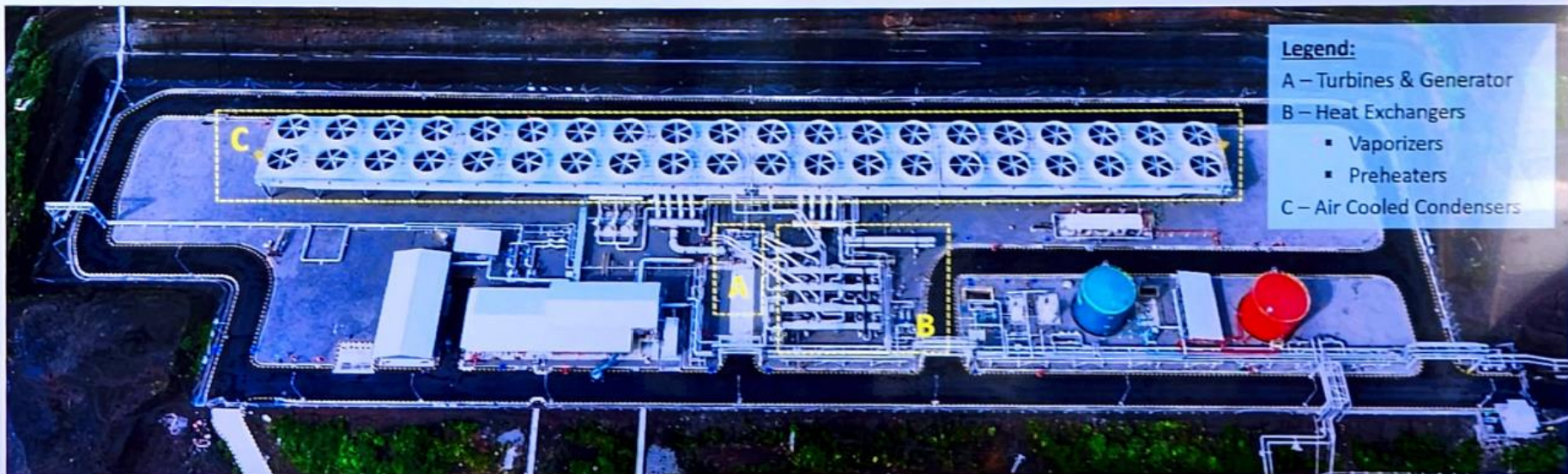
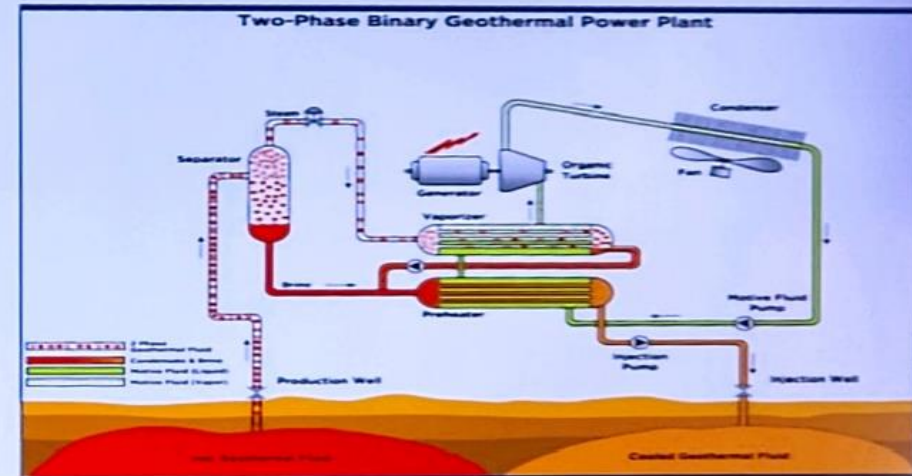
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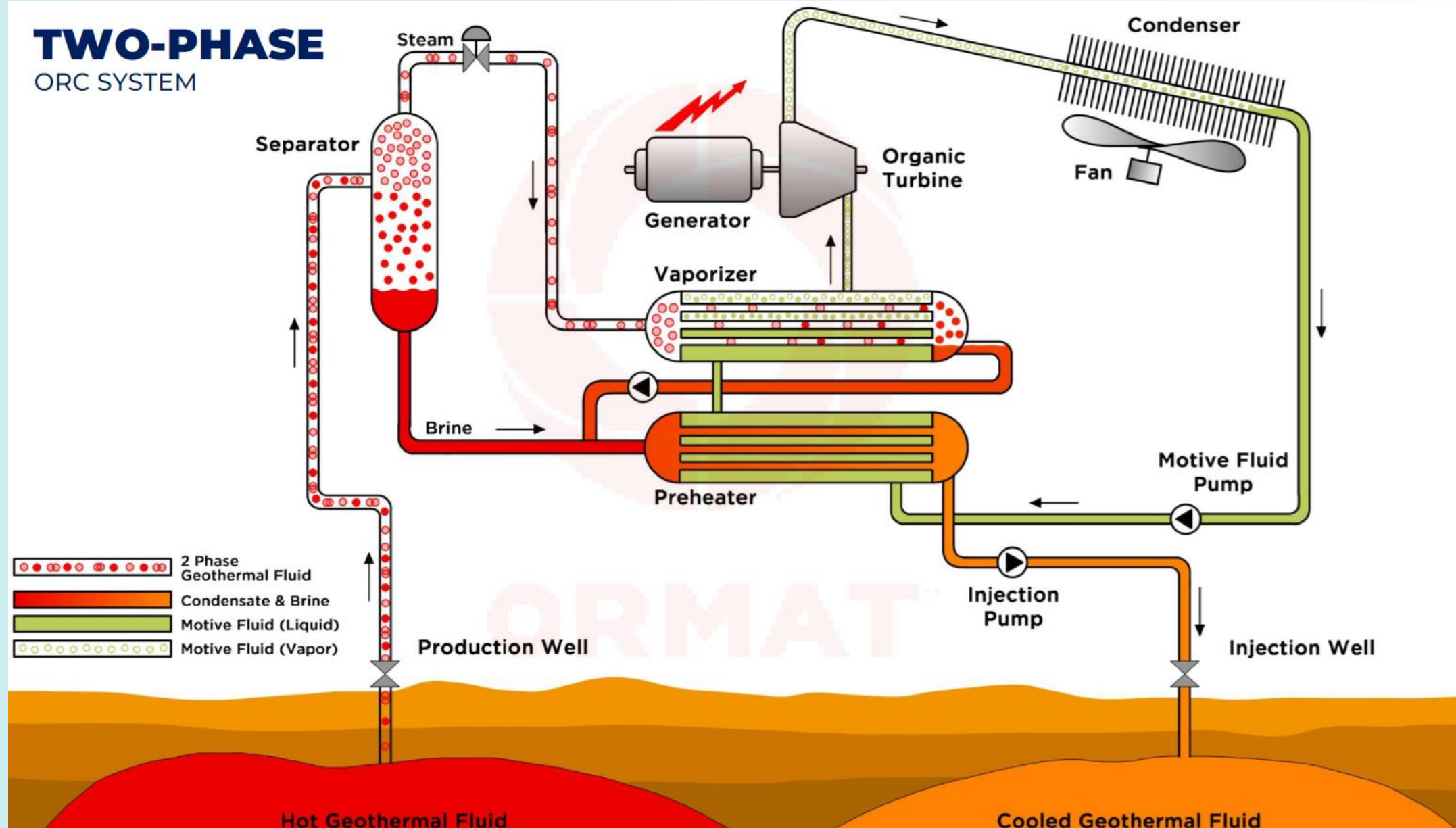
POWER PLANT TECHNOLOGY – TWO PHASE BINARY

Two-phase binary powerplant (OEC) is utilizing steam and brine from production wells to heat-up motive fluids, which then rotate turbines (2x17.5 MW) and generate the electricity.

Steam will go to vaporizers (4 units). Then, produced steam condensate mixed with brine and go to preheaters (2 units). Afterward, the mixed fluids will be reinjected to subsurface via injection well. After motive fluids (cyclopentane) rotate the turbine, it will be cooled down inside the ACC, before returning back to the OEC.

This system is 100% reinjection, which means more environmentally friendly and better to maintain the reservoir pressure (more sustain production).







CONVENTIONAL GEOTHERMAL

STEAM POWER PLANT

- **Application**

- Widely used for medium & high steam pressure resources
- Two-phase or dry steam resources
- Low NCG content

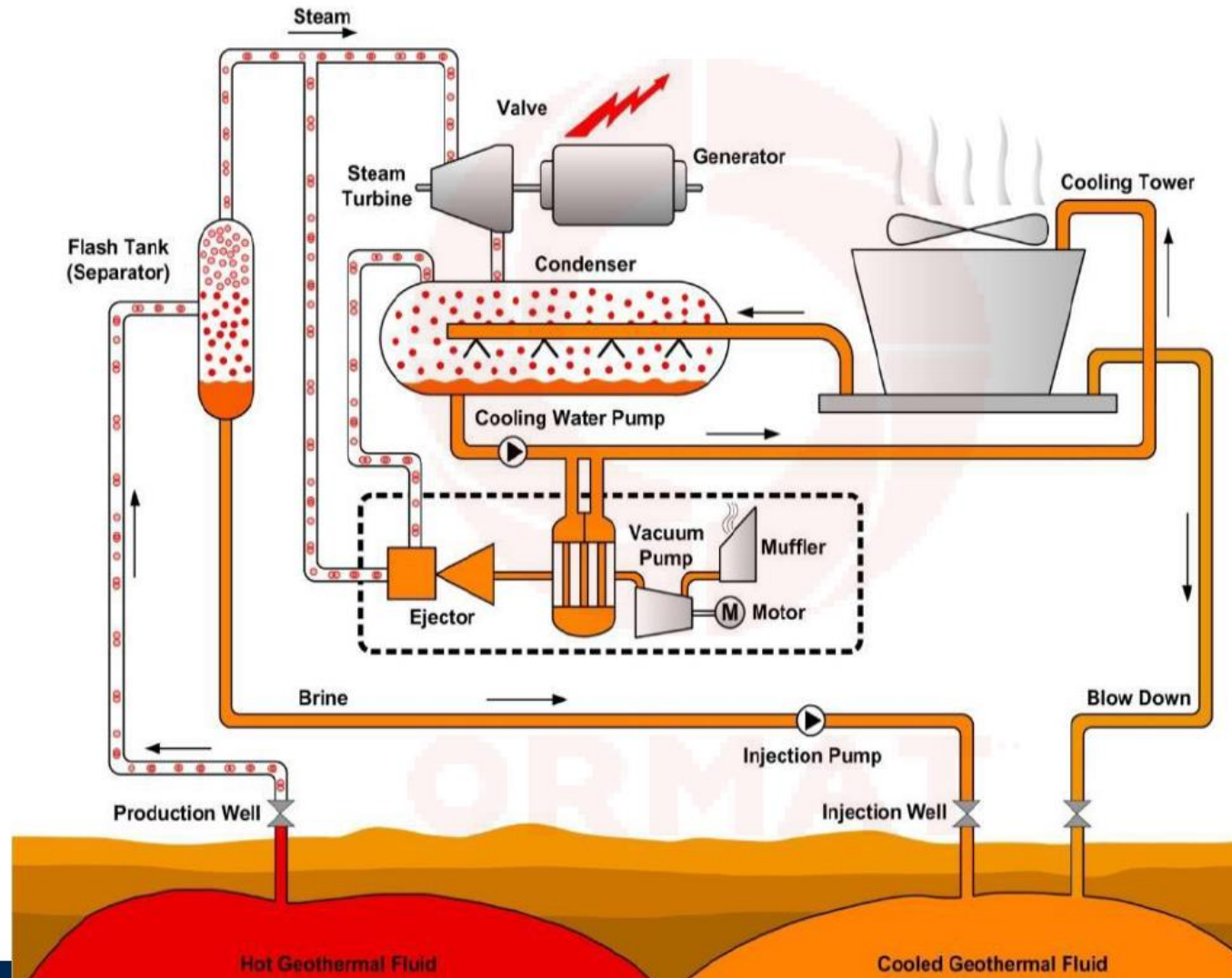
- **Benefits**

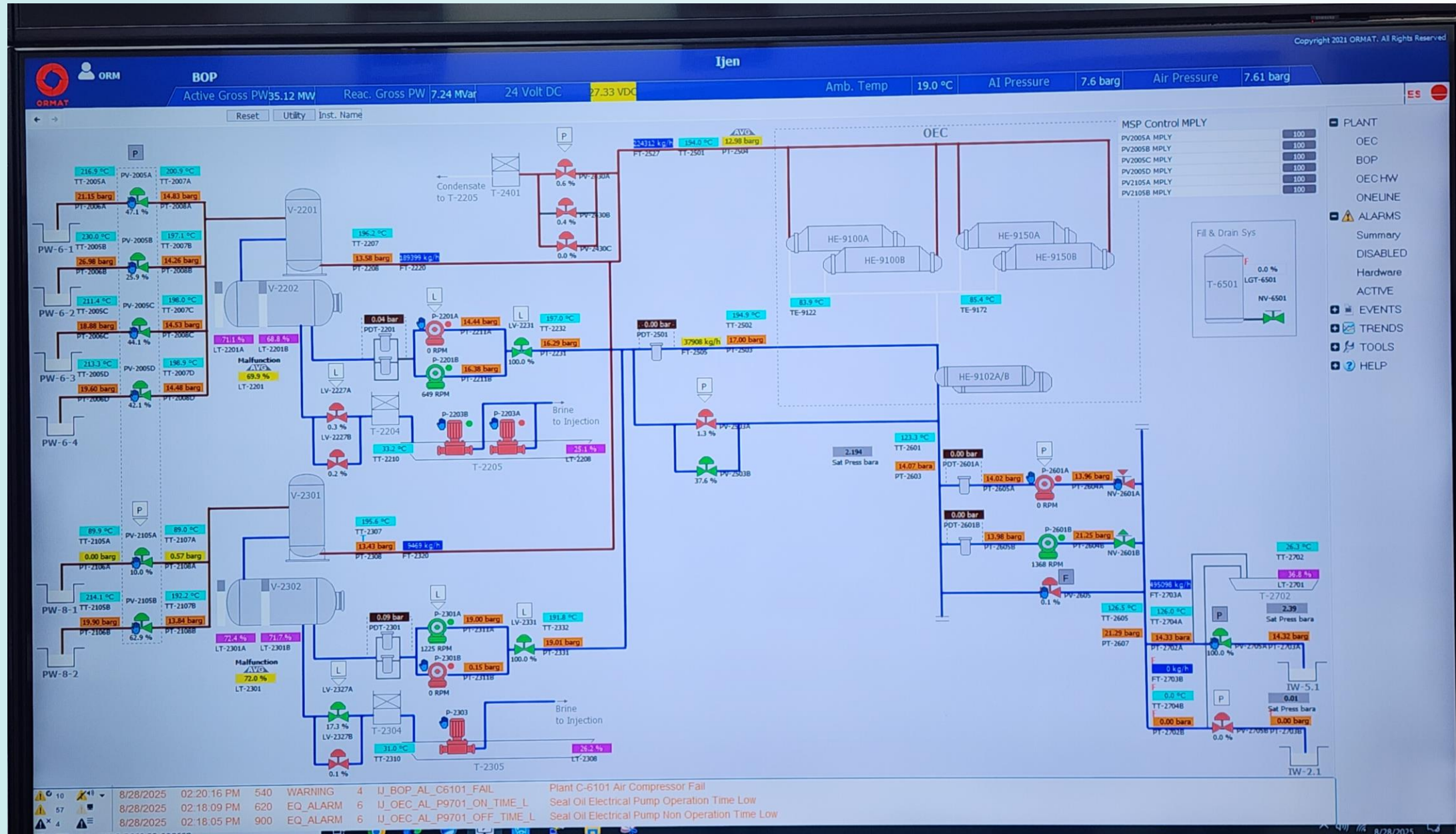
- High efficiency
- Large turbine capacity (over 100 MW)
- Robust, proven design
- Exhaust at low temperature (no relative metal expansion)

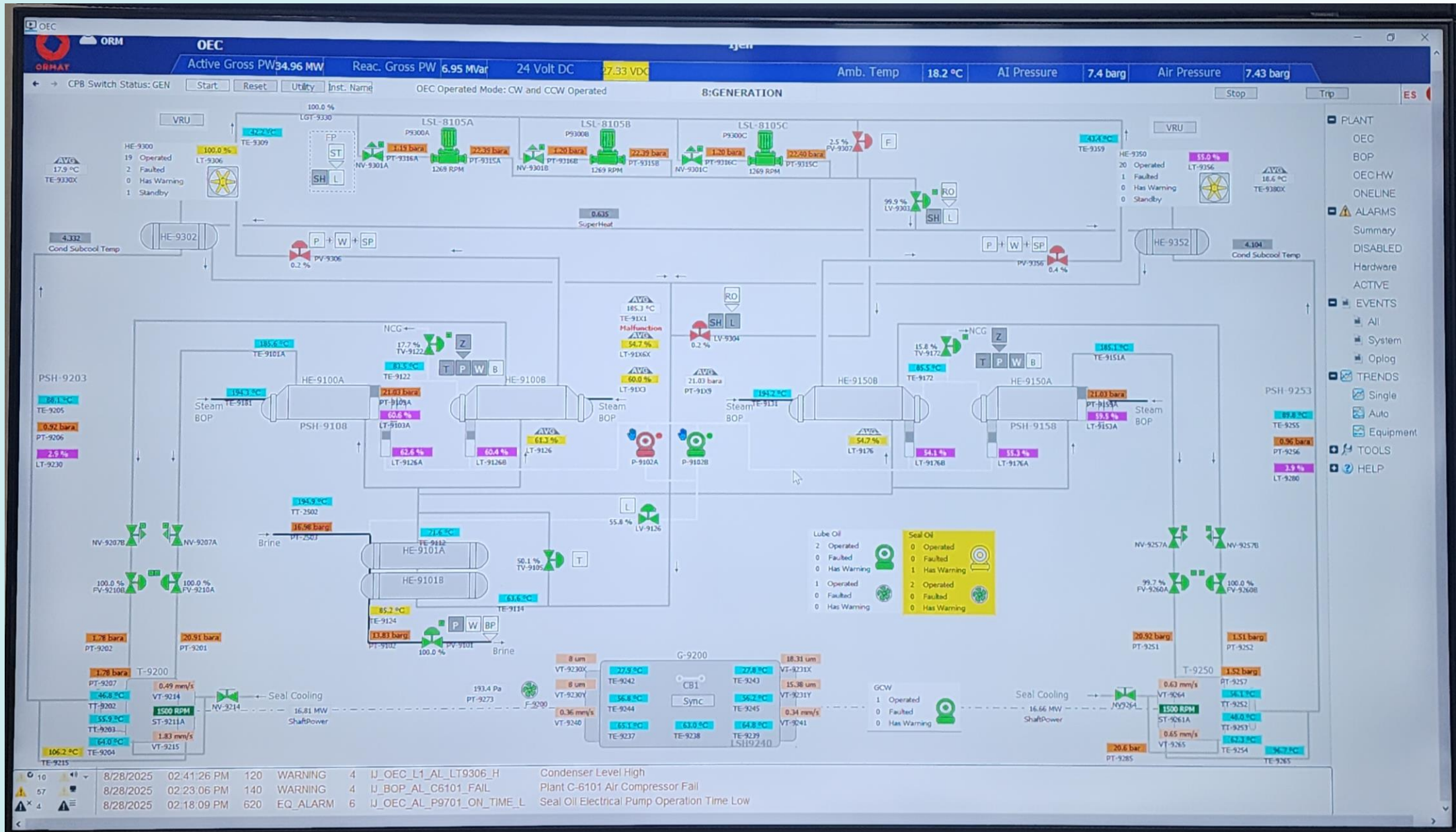
- **Disadvantages**

- Direct contact between geothermal steam and turbine blades
- High pressure steam causes excessive wetness in the turbine last stages
- Energy and/or steam consumption for disposal of NCG
- Less flexible to resource changes

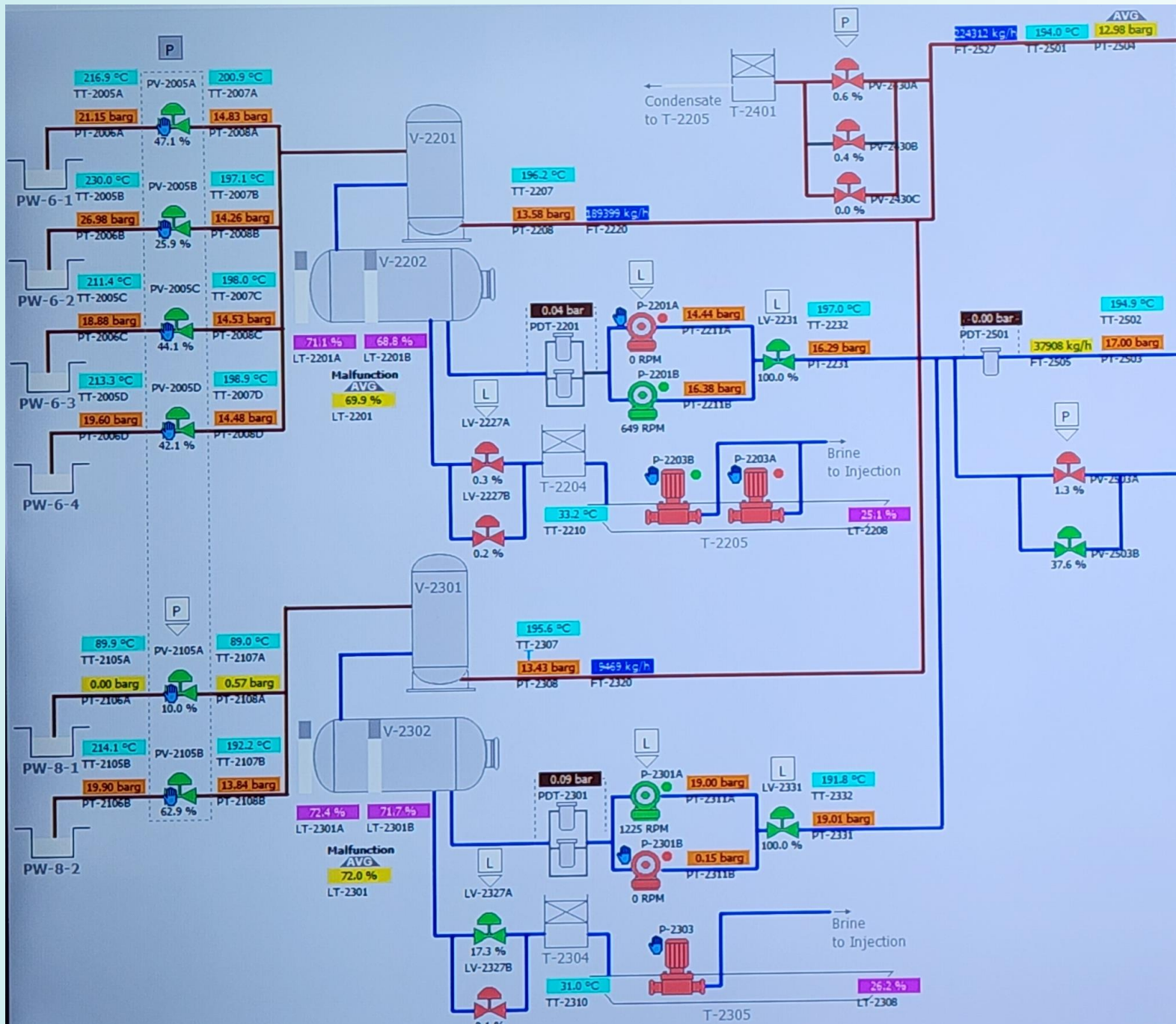
Geothermal Flash Power Plant







Steam Field



Steam before (inlet) vaporizer:

P (barg)	12.98
T (degC)	194
m (kg/h)	224,312

Brine before (inlet) preheater:

P (barg)	17
T (degC)	194.9
m (kg/h)	37,908



Ijen

Steam before (inlet) vaporizer:

P (barg)	12.98
T (degC)	194
m (kg/h)	224,312
m (kg/s)	62.3
m (ton/h)	224.312

Masih >> P turbin inlet
Kamojang (6.5 bara)

Misal langsung dihitung saja:

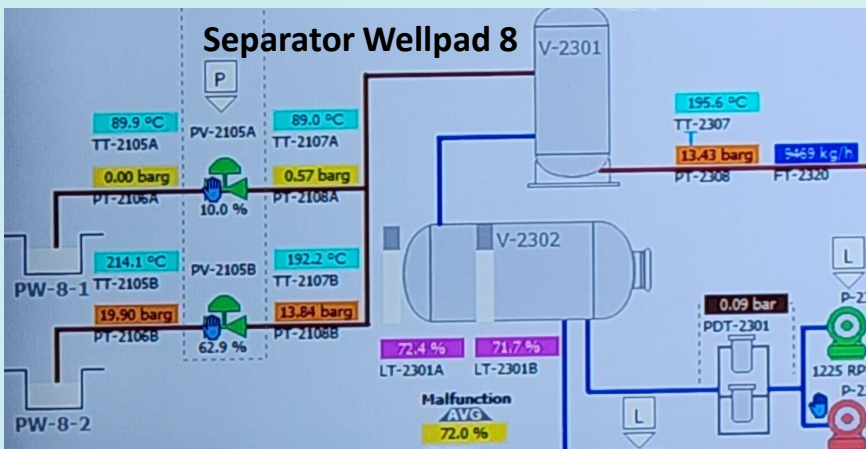
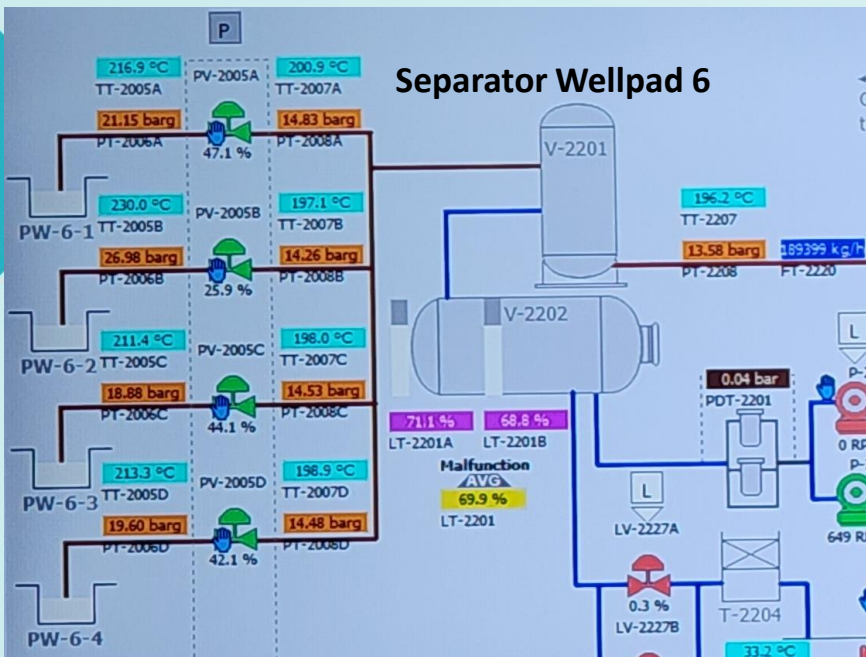
$$\frac{224.312 \text{ ton/h}}{7.17 \text{ ton/h/MW}} = 31.28 \text{ MW}$$

Belum termasuk
bottoming binary
dari brine

PLTP KAMOJANG Unit-1, 2 dan 3

Kamojang		Unit-1	Unit-2	Unit-3
Daya Listrik	MWe	30	55	55
Turbin Inlet Pressure	Bara	6,5	6,5	6,5
Tekanan Kondenser	Bara	0,1	0,1	0,1
Effisiensi Turbin		0,8	0,8	0,8
Entalpi uap @ inlet turbin. hg=h1	kJ/kg	2760	2760	2760
Entropi uap @ turbin. sg=s1	kJ/kg K	6,73	6,73	6,73
Entropi air @ Kondenser. sf2	kJ/kg K	0,65	0,65	0,65
Entropi uap @ Kondenser. sg2	kJ/kg K	8,15	8,15	8,15
Fraksi uap @ Kondenser. x2		0,81	0,81	0,81
Entalpi air @ Kondenser, hf2	kJ/kg	192	192	192
Entalpi uap @ Kondenser, hg2	kJ/kg	2584	2584	2584
Entalpi fluida @ Kondenser, hf2	kJ/kg	2132	2132	2132
Laju aliran massa uap, m	kg/s	60	110	110
Laju aliran massa uap, m	Ton/jam	215	394	394
Konsumsi Uap (SSC)	Ton/jam/MW	7,17	7,17	7,17
Total kebutuhan uap untuk Unit-1, 2 dan 3	Ton/jam	1004		
Steam Allowance	%	10		
Uap yang harus tersedia di lapangan (kebutuhan uap+steam allowance)	ton/jam	237	434	434
Total uap yang harus tersedia untuk Unit-1, 2 dan 3	Ton/jam	1104		





Reservoir (deep liquid):

Tres	290	degC
hl res	1289.80	kJ/kg

Separator Wellpad 6:

P	13.58	barg
T	196.2	degC
m uap	189,399	kg/h
m uap	52.61	kg/s
m uap	189.399	ton/h
hv	2787.92	kJ/kg
hl	823.78	kJ/kg
fraksi uap	0.24	
fraksi brine	0.76	
m brine	608,870	kg/h
m brine	169.13	kg/s
m brine	608.870	ton/h
m total	798,269	kg/h

Benchmark Turbin Uap Kamojang:

P inlet turbin	6.5	barg
T	196.2	degC
m uap	232,929	kg/h
m uap	64.70	kg/s
m uap	232.929	ton/h
hv	2759.60	kJ/kg
hl	684.22	kJ/kg
fraksi uap	0.29	
fraksi brine	0.71	
m brine	565,340	kg/h
m brine	157.04	kg/s
m brine	565.340	ton/h
m total	798,269	kg/h

Kapasitas MW Pembangkit Turbin Uap:

m uap Wellpad 6	232.929	ton/h
m uap Wellpad 8	11.599	ton/h
m uap Total	244.528	ton/h
Konsumsi Uap (SSC)	7.17	ton/h/MW
Kapasitas PLTP	34.10	MW

Belum termasuk
bottoming binary
dari brine

Separator Wellpad 8:

P	13.43	barg
T	195.6	degC
m uap	9,469	kg/h
m uap	2.63	kg/s
m uap	9.469	ton/h
hv	2787.56	kJ/kg
hl	821.47	kJ/kg
fraksi uap	0.24	
fraksi brine	0.76	
m brine	30,283	kg/h
m brine	8.41	kg/s
m brine	30.283	ton/h
m total	39,752	kg/h

Benchmark Turbin Uap Kamojang:

P inlet turbin	6.5	barg
T	195.6	degC
m uap	11,599	kg/h
m uap	3.22	kg/s
m uap	11.599	ton/h
hv	2759.60	kJ/kg
hl	684.22	kJ/kg
fraksi uap	0.29	
fraksi brine	0.71	
m brine	28,153	kg/h
m brine	7.82	kg/s
m brine	28.153	ton/h
m total	39,752	kg/h



Wellpad 6:

P inlet turbin	6.5	barg
T	196.2	degC
m uap	232,929	kg/h
m uap	64.70	kg/s
m uap	232.929	ton/h
hv	2759.60	kJ/kg
hl	684.22	kJ/kg
fraksi uap	0.29	
fraksi brine	0.71	
m brine	565,340	kg/h
m brine	157.04	kg/s
m brine	565.340	ton/h
m total	798,269	kg/h

Wellpad 8:

P inlet turbin	6.5	barg
T	195.6	degC
m uap	11,599	kg/h
m uap	3.22	kg/s
m uap	11.599	ton/h
hv	2759.60	kJ/kg
hl	684.22	kJ/kg
fraksi uap	0.29	
fraksi brine	0.71	
m brine	28,153	kg/h
m brine	7.82	kg/s
m brine	28.153	ton/h
m total	39,752	kg/h

Kapasitas MW Pembangkit Turbin Uap:

m uap Wellpad 6	232.929	ton/h
m uap Wellpad 8	11.599	ton/h
m uap Total	244.528	ton/h
Konsumsi Uap (SSC)	7.17	ton/h/MW
Kapasitas PLTP	34.10	MW

Belum termasuk
bottoming binary
dari brine

Kapasitas MW Pembangkit Binary:

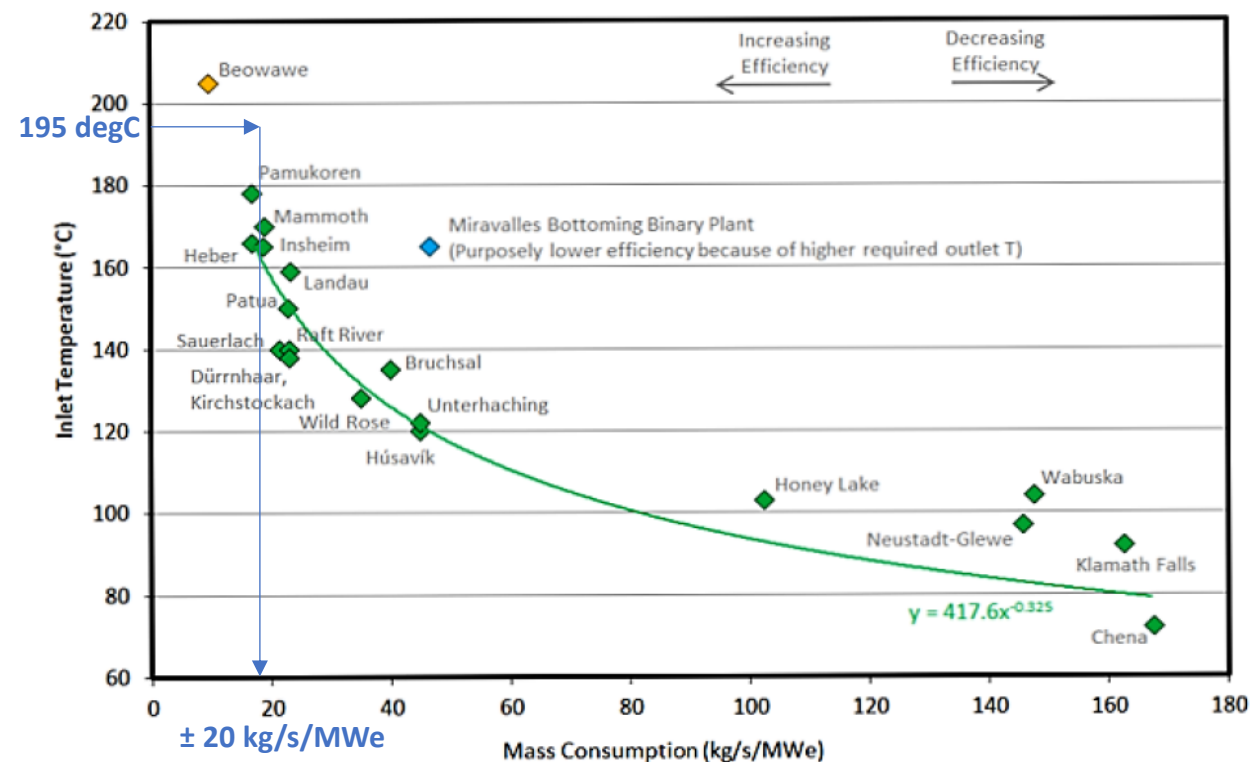
m brine Wellpad 6	565.340	ton/h
m brine Wellpad 8	28.153	ton/h
m brine total	593.493	ton/h
m brine total	164.859	kg/s
P	6.5	barg
T	195.6	degC
hl	832.61	kJ/kg
Mass Consumption	20	kg/s/MWe
Kapasitas PLTP	8.24	MWe

Kapasitas MW Total:

Single Flash	34.10	MW
Binary	8.24	MW
Combined Cycle	42.35	MW

MASS CONSUMPTION

Temperature vs Flow rate to generate 1 MW



Optimistic

Reservoir (deep liquid):											
Tres	290	degC	290-324 deg C (liq dominated reservoir)								
hl res	1289.80	kJ/kg									
Separator Wellpad 6:			Benchmark Turbin Uap Kamojang:			Kapasitas MW Pembangkit Turbin Uap:					
P	13.58	barg	P inlet turbin	6.5	barg	m uap Wellpad 6	232.929	ton/h			
T	196.2	degC	T	196.2	degC	m uap Wellpad 8	11.599	ton/h			
m uap	189,399	kg/h	m uap	232,929	kg/h	m uap Total	244.528	ton/h			
m uap	52.61	kg/s	m uap	64.70	kg/s	Konsumsi Uap (SSC)	7.17	ton/h/MW			
m uap	189,399	ton/h	m uap	232,929	ton/h	Kapasitas PLTP	34.10	MW			
hv	2787.92	kJ/kg	hv	2759.60	kJ/kg						
hl	823.78	kJ/kg	hl	684.22	kJ/kg						
fraksi uap	0.24		fraksi uap	0.29							
fraksi brine	0.76		fraksi brine	0.71							
m brine	608,870	kg/h	m brine	565,340	kg/h	Kapasitas MW Pembangkit Binary:					
m brine	169.13	kg/s	m brine	157.04	kg/s	m brine Wellpad 6	565.340	ton/h			
m brine	608.870	ton/h	m brine	565,340	ton/h	m brine Wellpad 8	28.153	ton/h			
m total	798,269	kg/h	m total	798,269	kg/h	m brine total	593.493	ton/h			
						m brine total	164.859	kg/s			
Separator Wellpad 8:			Benchmark Turbin Uap Kamojang:			P	6.5	barg			
P	13.43	barg	P inlet turbin	6.5	barg	T	195.6	degC			
T	195.6	degC	T	195.6	degC	hl	832.61	kJ/kg			
m uap	9,469	kg/h	m uap	11,599	kg/h	Mass Consumption	20	kg/s/MWe			
m uap	2.63	kg/s	m uap	3.22	kg/s	Kapasitas PLTP	8.24	MWe			
m uap	9,469	ton/h	m uap	11,599	ton/h						
hv	2787.56	kJ/kg	hv	2759.60	kJ/kg						
hl	821.47	kJ/kg	hl	684.22	kJ/kg						
fraksi uap	0.24		fraksi uap	0.29							
fraksi brine	0.76		fraksi brine	0.71							
m brine	30,283	kg/h	m brine	28,153	kg/h	Kapasitas MW Total:					
m brine	8.41	kg/s	m brine	7.82	kg/s	Single Flash	34.10	MW			
m brine	30,283	ton/h	m brine	28,153	ton/h	Binary	8.24	MW			
m total	39,752	kg/h	m total	39,752	kg/h	Combined Cycle	42.35	MW			

Kapasitas MW Total:

Single Flash	34.10	MW
Binary	8.24	MW
Combined Cycle	42.35	MW

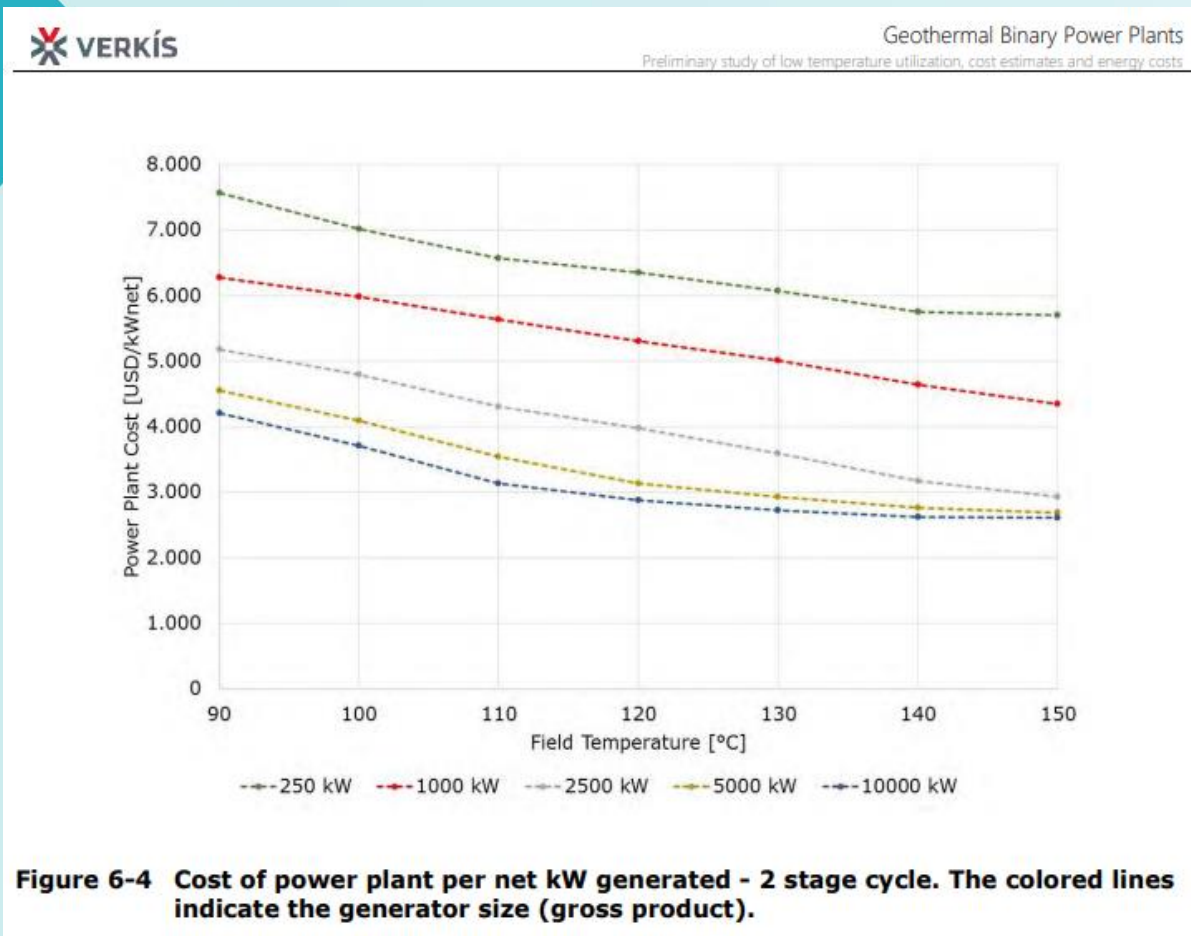
Pessimistic

Reservoir (deep liquid):									
Tres	324	degC	290-324 deg C (liq dominated reservoir)						
hl res	1487.03	kJ/kg							
Separator Wellpad 6:			Benchmark Turbin Uap Kamojang:			Kapasitas MW Pembangkit Turbin Uap:			
P	13.58	barg	P inlet turbin	6.5	barg	m uap Wellpad 6	216.965	ton/h	
T	196.2	degC	T	196.2	degC	m uap Wellpad 8	10.820	ton/h	
m uap	189,399	kg/h	m uap	216,965	kg/h	m uap Total	227.786	ton/h	
m uap	52.61	kg/s	m uap	60.27	kg/s	Konsumsi Uap (SSC)	7.17	ton/h/MW	
m uap	189,399	ton/h	m uap	216,965	ton/h	Kapasitas PLTP	31.77	MW	
hv	2787.92	kJ/kg	hv	2759.60	kJ/kg				
hl	823.78	kJ/kg	hl	684.22	kJ/kg				
fraksi uap	0.34		fraksi uap	0.39					
fraksi brine	0.66		fraksi brine	0.61					
m brine	371,485	kg/h	m brine	343,919	kg/h	Kapasitas MW Pembangkit Binary:			
m brine	103.19	kg/s	m brine	95.53	kg/s	m brine Wellpad 6	343.919	ton/h	
m brine	371,485	ton/h	m brine	343,919	ton/h	m brine Wellpad 8	17.152	ton/h	
m total	560,884	kg/h	m total	560,884	kg/h	m brine total	361.070	ton/h	
						m brine total	100.297	kg/s	
Separator Wellpad 8:			Benchmark Turbin Uap Kamojang:			P	6.5	barg	
P	13.43	barg	P inlet turbin	6.5	barg	T	195.6	degC	
T	195.6	degC	T	195.6	degC	hl	832.61	kJ/kg	
m uap	9,469	kg/h	m uap	10,820	kg/h	Mass Consumption	20	kg/s/MWe	
m uap	2.63	kg/s	m uap	3.01	kg/s	Kapasitas PLTP	5.01	MWe	
m uap	9,469	ton/h	m uap	10,820	ton/h				
hv	2787.56	kJ/kg	hv	2759.60	kJ/kg				
hl	821.47	kJ/kg	hl	684.22	kJ/kg	Kapasitas MW Total:			
fraksi uap	0.34		fraksi uap	0.39		Single Flash	31.77	MW	
fraksi brine	0.66		fraksi brine	0.61		Binary	5.01	MW	
m brine	18,503	kg/h	m brine	17,152	kg/h	Combined Cycle	36.78	MW	
m brine	5.14	kg/s	m brine	4.76	kg/s				
m brine	18,503	ton/h	m brine	17,152	ton/h				
m total	27,972	kg/h	m total	27,972	kg/h				

Kapasitas MW Total:

Single Flash	31.77	MW
Binary	5.01	MW
Combined Cycle	36.78	MW

Binary



Verkis, 2014

Kapasitas	5	MW
Temp	>150	degC
Cost	2.8	USD/MW
Escalation	2%	
Current Cost	3.35	USD/MW

Single Flash

Parameters	110 MWe	55 MWe	20 MWe	10 MWe
Power Generation Facilities (MS/MW)	1.5	1.7	1.9	2.0
Pwr. Plant Cost (Condensing Turbine)	169,400	93,170	37,268	20,497

Wahjosoedibjo & Hasan, 2018

Kapasitas	20	MW
Cost	1.9	USD/MW
Escalation	2%	
Current Cost	2.10	USD/MW



TERIMA KASIH

The New PLN 4.0 Unleashing Energy and Beyond

